Rate-controllable PPG synthesis for a low-cost simulator: rhythm-scaffolded GAN and rectified flow under pre-registered evaluation

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Abstract

Bench simulators of the photoplethysmogram (PPG) let developers test pulse-rate and rhythm algorithms without recruiting subjects, but the parametric pulses they play back are easy to tell apart from recorded PPG. We built a Raspberry Pi 4 simulator that drives red and infrared LEDs from two 12-bit converters, together with two generators that construct the beat sequence from the commanded rate and irregularity and learn only pulse morphology and texture: PhysioMix, a conditional generative adversarial network (0.16 M parameters), and PhysioFlow, a conditional rectified-flow U-Net (6.9 M parameters). Both were trained on six public datasets with subject-disjoint splits and selected on validation data under criteria fixed before training. Their mean absolute rate error was below 1 bpm from 40 to 180 bpm. On subjects never used for training, a subject-grouped classifier two-sample test rated PhysioFlow 50–91% indistinguishable from real windows on four finger and intensive-care datasets, against 7–45% for PhysioMix and 0% for the parametric pulse; a frozen PPG foundation model ranked the generators in the same order (Spearman $\rho = 0.88$). Neither model transferred to everyday wrist PPG from an unseen device (below 1%). Training with that device's data raised PhysioFlow to 33% on held-out wearers, but its validation gain missed the pre-registered margin, so the deployed default remained PhysioMix. On the Pi, the converters were updated at 500 Hz without overruns; the optical path is not yet validated.

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1. Introduction

Photoplethysmography (PPG) measures changes in tissue blood volume optically. It is the sensing principle behind pulse oximeters, smartwatches and many bedside and ambulatory monitors [1, 2]. Algorithms that run on the PPG estimate pulse rate, detect atrial fibrillation, extract respiratory rate and grade signal quality [3, 4, 5], and denoising methods are being developed for the motion-corrupted signals of wearables [6, 7]. Before such algorithms, and the optical front-ends that feed them, can be trusted, they have to be tested on signals whose properties are known.

Two kinds of test signal are in use. Recorded databases contain real physiology, and a beat detector can be scored against them [8], but they cannot be varied: rate, rhythm and recording conditions are fixed by the recording, and a file cannot exercise an optical sensor. Bench simulators reverse the trade-off. Commercial testers place an optical module in front of the device under test and modulate light with a pulse whose rate, amplitude and timing features are set by the operator; the WhaleTeq AECG100, for example, covers 10–300 bpm with PPG, sine, triangle and square waveforms and can also play back user-supplied data [9]. Such functional testers show that a device responds, but they do not establish SpO_2 accuracy, which is assessed in controlled studies on human subjects [10]. Academic bench simulators follow the same pattern. Pittella et al. drove multi-wavelength LEDs to reproduce database PPG in front of an earbud sensor [11], and Sološenko et al. built an LED-based simulator driven by beat-to-beat interval series to reproduce ventricular tachycardia for wrist-worn devices [12].

The waveform is the weak point of these systems. Parametric models built from sums of Gaussians [13, 14], stochastic pulse models [15], rhythm models of atrial fibrillation [16, 17] and haemodynamic pulse-wave databases [18] give exact control, but their pulses repeat more regularly than real ones [19]. In our evaluation, a small convolutional classifier separated the simulator's Gaussian pulse from real windows of unseen subjects in all six datasets. Playing back recordings gives realism without control. Deep generative models fill part of this gap. Generative adversarial networks (GANs) [20] have produced PPG for data augmentation [21, 22], and diffusion and flow-matching models now generate electrocardiograms conditioned on diagnostic statements

[23, 24] or translate PPG into ECG [25, 26]. These models were not built to follow an operator's rate command, and their realism is often judged on a few examples, with a single distance or on test data that were seen during development [27, 28].

A simulator places four requirements on a generator. The rate and rhythm must follow the command exactly, including outside the range that dominates the training data. The waveform should be realistic in the recording context the user selects, such as a fingertip at rest, a patient in intensive care or a wrist during exercise. Inference has to fit a small Linux computer that also drives the converters in real time. Finally, users need to know where the model is realistic and where it is not, which requires evaluation on unseen subjects with metrics that a generator cannot easily satisfy by accident. This last requirement also concerns model selection. When the same validation subjects are reused over many development rounds, validation scores become optimistic, and gains reported on them may not survive a fresh test [29, 30]. Fixing the analysis before the data are seen is the standard remedy in other fields [31], and we apply it here to the development of signal generators.

This paper describes how we addressed these requirements and where we fell short. Its contributions are:

- an open, low-cost simulator built around a Raspberry Pi 4 with two 12-bit converters, an LED current-sink driver, an OPT101 loopback receiver and a printed dark chamber, whose real-time converter output was measured on the device (Section 2);
- two conditional generators, PhysioMix (a GAN) and PhysioFlow (a rectified-flow model), that construct the beat sequence from the rate and irregularity commands and learn only the waveform around it (Section 3);
- a selection protocol in which every decision rule was written and hashed before training and applied to validation data only, complemented by realism measurements on unseen subjects: fidelity and coverage, a subject-grouped classifier two-sample test, a frozen PPG foundation model, recording-context classification, memorisation screening and a fresh test on everyday wrist PPG (Section 4);
- results that include negative findings: exact rate control in both models, a clear realism advantage of rectified flow on finger and intensive-care data, the failure of both models on everyday wrist PPG from an

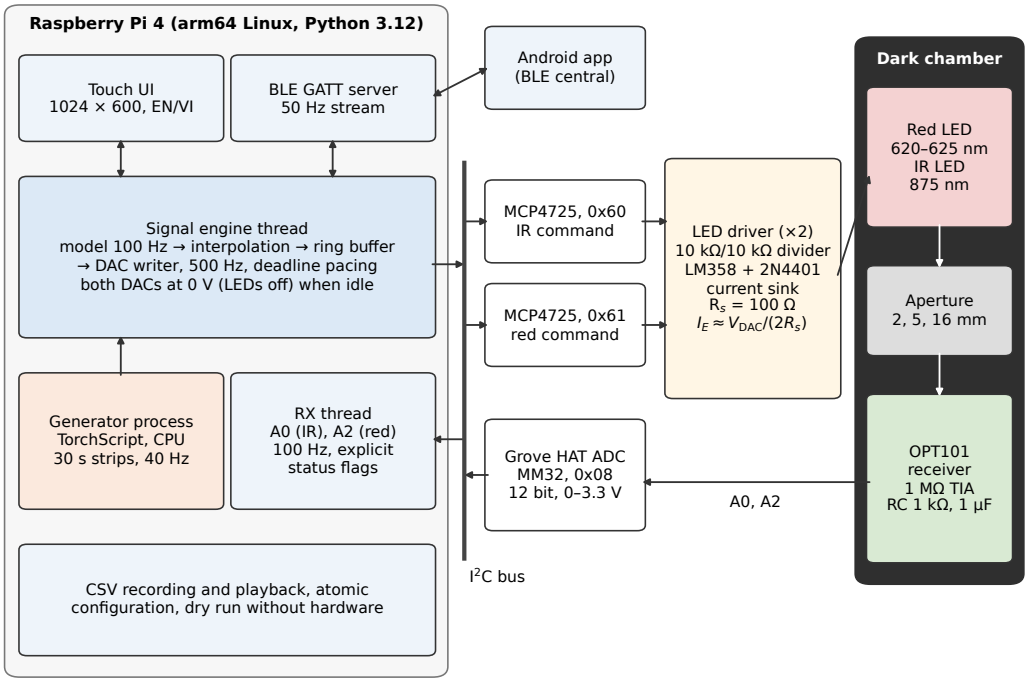


Figure 1: Architecture of the simulator. Software threads and the generator process run on the Raspberry Pi; the two converters and the receiver ADC share one I^2C bus. The LED driver and the receiver boards are described in Section 2.1; the optical path has not yet been validated (Section 5.1). Diagram drawn with Matplotlib from code written with the help of Claude (Anthropic).

unseen device, and two cases in which the pre-registered rules kept the deployed model unchanged (Section 5).

2. Simulator platform

2.1. Hardware

Fig. 1 shows the architecture. A Raspberry Pi 4 Model B runs the application and shares one I^2C bus between two MCP4725 12-bit converters [32] at addresses 0x60 (infrared command) and 0x61 (red command) and the analogue-to-digital converter of a Grove Base HAT, an MM32 microcontroller at address 0x08 that digitises 0–3.3 V with 12 bits. The converters are supplied from the 3.3 V rail, measured at 3.28 V on our unit, so one code step corresponds to 0.80 mV.

Each converter output drives one LED through a voltage-controlled current sink (Fig. 2a). A 10 kΩ/10 kΩ divider halves the command, and one half of an LM358 operational amplifier drives a 2N4401 transistor so that the voltage across the emitter resistor $R_s = 100\,\Omega$ follows the divided command. The nominal emitter current is therefore

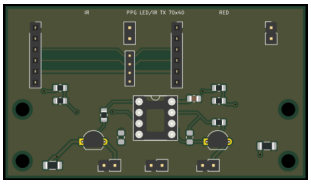
$$I_E \approx \frac{V_{\mathrm{DAC}}}{2R_s}, \tag{1}$$

about 16.4 mA at full scale; the LED current is smaller by the base current. The LEDs are 3 mm parts with a dominant wavelength of 620–625 nm (red) and a peak wavelength of 875 nm (infrared). Their anodes connect to the 5 V rail, which supplies only the amplifier and the LEDs. Starting, stopping and closing the application park both converters at code 0, which turns the LEDs off.

Light reaches one of two interchangeable receiver boards. The default board carries two OPT101 photodiode amplifiers [33] with their internal 1 MΩ feedback resistor and a 1 kΩ/1 µF output filter (nominal cut-off 159 Hz) ahead of the HAT inputs A0 (infrared) and A2 (red) (Fig. 2b). The alternative board uses BPW34 photodiodes with OPA333 transimpedance amplifiers (1 MΩ in parallel with 22 pF); integrated front-ends with ambient-light cancellation [34] would be the natural next step. A printed dark chamber separates the red and infrared lanes, holds the LEDs on sliding carriers and accepts apertures of 2, 5 and 16 mm or a blank. The boards were designed in KiCad and pass its electrical-rule and design-rule checks. Their electrical and optical behaviour has not been measured; the bench capture of Section 5.1 used an earlier assembly with an OPT101 module on A0 only.

2.2. Software and real-time output

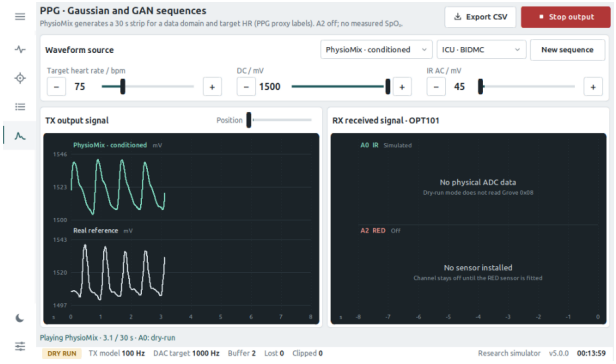
The application is written in Python 3.12. Its touch interface, built with CustomTkinter, adapts from 1024 × 600 panels to desktop displays and offers English and Vietnamese (Fig. 2c). A single signal-engine thread evaluates the parametric model of Section 2.3 at 100 Hz, interpolates linearly to the converter rate, stores the samples in a bounded ring buffer and writes both converters once per tick, infrared first. The converter rate can be 100, 200, 500 or 1000 Hz; on the Pi the default is 500 Hz for the reason given in Section 5.1. Writes are paced against absolute deadlines: the writer sleeps only for the time left before the next deadline and yields without sleeping when it



(a) LED driver, 70 × 40 mm



(b) OPT101 receiver, 70 × 20 mm



(c) Generator page, laptop dry run, 1024 × 600

Figure 2: (a, b) KiCad renders of the LED driver and OPT101 receiver boards, shown with empty sockets. (c) Generator page of the application in dry-run mode on a laptop at 1024×600: a PhysioMix strip prompted for BIDMC at 75 bpm (top trace) above a real training strip of the same dataset; the receiver card on the right reports that no ADC data exist instead of drawing a substitute.

is late. A lock serialises all converter writes, because the driver library shares one transmit buffer between instances.

Learned generators never run on the engine or interface threads. The interface starts a worker process that loads a TorchScript generator, produces one 30 s strip at 40 Hz and returns it as JSON. A strip $x \in [0, 1]$ is mapped to commanded voltages

$$v_{\mathrm{IR}}(t) = V_{\mathrm{DC}} + A_{\mathrm{IR}}\, x(t), \qquad v_{\mathrm{red}}(t) = V_{\mathrm{DC}} + 0.48\, A_{\mathrm{IR}}\, x(t), \tag{2}$$

with $V_{\mathrm{DC}} = 1.5\,\mathrm{V}$ and $A_{\mathrm{IR}} = 45\,\mathrm{mV}$ by default, both adjustable. The fixed red-to-infrared ratio is a nominal mapping, not a learned optical relation. The strip is interpolated to 100 Hz and then to the converter rate; interpolation adds no bandwidth. Playback ends after 30 s or when the user leaves the page, and both converters then return to 0 V.

A receiver thread samples A0 and A2 at 100 Hz and reports explicit states (live, stale, disabled, dry run and full scale) instead of drawing values when a sensor is absent. A Bluetooth Low Energy GATT server lets an Android application set parameters and receive the waveform at 50 Hz with a 5 Hz status message. Recordings store the commanded samples at 100 Hz with timestamps, and the configuration is written atomically. A dry-run mode replaces all hardware access, so the complete application and its automated test suite of more than 800 tests run on a laptop.

2.3. Parametric waveform engine

The conventional mode reproduces the PPG and SpO_2 controls of commercial testers [9]. One cardiac cycle is generated from the phase $\theta \in [0, 1)$ as

$$p(\theta) = \frac{E(\theta)\,\big[g(\theta;\mu_s,\sigma_s) + a_d\,g(\theta;\mu_d,\sigma_d)\big]\,\big[1 - d\,g(\theta;\mu_n,\sigma_n)\big]}{\max_\theta(\cdot)}, \qquad (3)$$

where g is a unit-height Gaussian, E a smooth taper that joins consecutive cycles, $a_d = 0.4$ the diastolic weight and $d = 0.25$ the depth of a local notch attenuation. The default centres are $\mu_s, \mu_n, \mu_d = 0.15, 0.30, 0.40$ and the widths $\sigma_s, \sigma_n, \sigma_d = 0.055, 0.02, 0.10$ of a cycle. Feature times are specified at 60 bpm and scale with the cycle. Each channel has its own DC level and AC amplitude, and the perfusion index is $\mathrm{PI} = 100\,A_{\mathrm{IR}}/V_{\mathrm{DC,IR}}$. A target saturation sets the ratio of ratios [35]

$$R = \frac{A - \mathrm{SpO_2}}{B}, \qquad A_{\mathrm{red}} = R\,A_{\mathrm{IR}}\,\frac{V_{\mathrm{DC,red}}}{V_{\mathrm{DC,IR}}}, \qquad (4)$$

with the empirical linear calibration $A = 110$, $B = 25$ that the commercial tester uses by default [9]. Respiration modulates baseline, amplitude and beat timing at 1–150 breaths/min with an adjustable inspiration-to-expiration ratio and optional apnoea [3], and the heart rate spans 10–300 bpm. Because R is set rather than measured, this mode commands an optical ratio; it does not turn the simulator into an SpO_2 calibrator. The pulse of Eq. (3), rendered at the commanded rate and passed through the data view of Section 3.1, is the Gaussian baseline of the evaluation.

3. Rhythm-scaffolded generators

3.1. Data view and condition vector

All learned models work on the same view of the data. Each 30 s window is resampled to 40 Hz (1200 samples), filtered by a zero-phase 0.5–8 Hz Butterworth band-pass and scaled to $[0, 1]$ by its own minimum and maximum. The view keeps the pulse and its harmonics up to 8 Hz but removes the slow respiratory baseline and the absolute amplitude, so the generators learn normalised shape and timing only.

The condition vector $c \in \mathbb{R}^{11}$ contains a one-hot code for the six training datasets, two flags that state whether the window's rate and irregularity labels are known, and three values standardised with training statistics: pulse

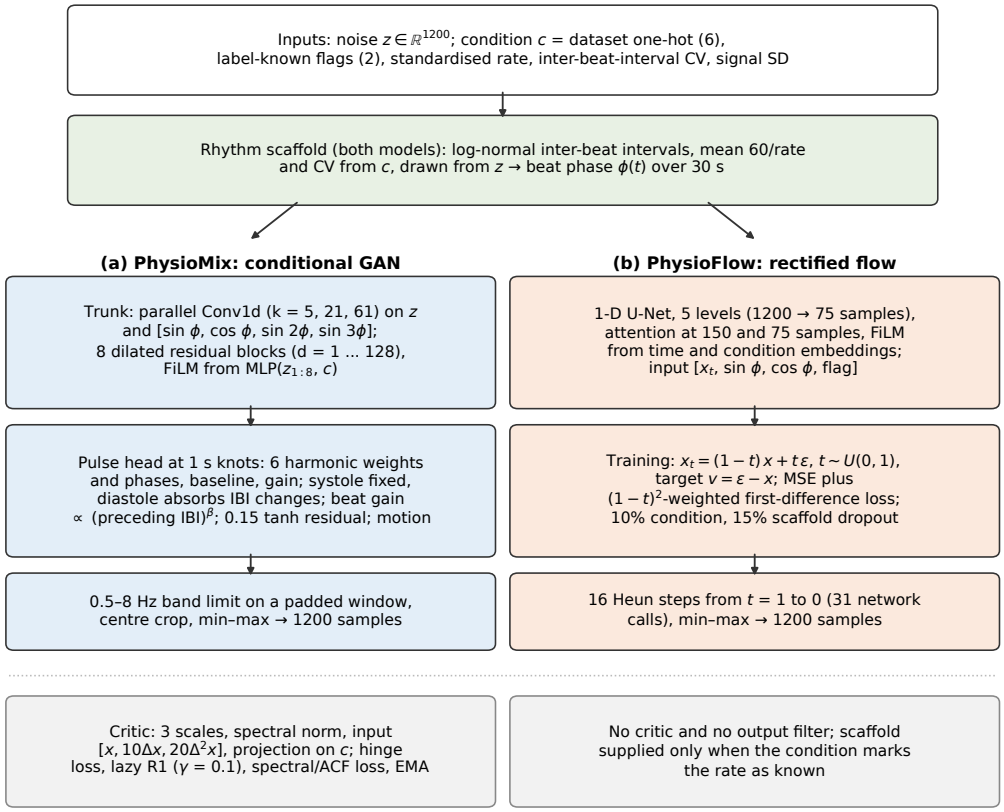


Figure 3: The two generators share the inputs and the rhythm scaffold, which fixes the beat times from the rate and irregularity commands. (a) PhysioMix shapes each beat with a convolutional trunk and harmonic pulse head and is trained against a multi-scale critic. (b) PhysioFlow learns a velocity field with a one-dimensional U-Net and generates by integrating it from noise. Diagram drawn with Matplotlib from code written with the help of Claude (Anthropic).

rate, coefficient of variation (CV) of the inter-beat intervals (IBIs) and the standard deviation of the window. Rate labels come from the agreement of spectral, autocorrelation and beat-interval estimates; where an ECG is available, the ECG rate is used when the PPG estimate agrees with it, and the label is masked otherwise. On the finger dataset described below, the median absolute difference between the PPG and ECG rates was 0.47 bpm. Wrist PPG during exercise is dominated by motion: its PPG rate differed from the ECG rate by a median of 52 bpm, and 86% of its rate labels were masked. Masked labels are not imputed; the corresponding flag is set to zero.

3.2. Rhythm scaffold

Both generators construct the beat sequence instead of learning it. Given a rate command r (bpm) and an irregularity command σ (the IBI CV, clipped at 0.35), the IBIs are drawn as

$$\tau_k = \frac{\exp(\sigma u_k - \sigma^2/2)}{f}, \qquad f = \frac{r}{60}, \quad u_k \sim \mathcal{N}(0,1), \tag{5}$$

so that $\mathbb{E}[\tau_k] = 1/f$ for any σ. The u_k are read from the generator's noise vector, and a random fraction of the first interval sets the phase at the start of the window. Cumulative sums give beat onsets (PhysioMix) or systolic peak times (PhysioFlow), and the beat phase $\phi(t) = 2\pi\,[k(t) + \delta(t)]$ combines the index $k(t)$ of the current beat with the elapsed fraction $\delta(t)$ of its interval. Because the networks see only $\phi(t)$, the rate cannot drift away from the command, and commands outside the label range of the training data remain valid. The log-normal law describes the dispersion of the intervals but not their sequential structure, so it is not a validated model of atrial fibrillation.

3.3. PhysioMix: a conditional GAN

PhysioMix (Fig. 3a) turns the scaffold into a waveform with 162,216 parameters. A trunk of three parallel one-dimensional convolutions (kernels of 5, 21 and 61 samples) reads the noise and four carriers ($\sin\phi$, $\cos\phi$, $\sin 2\phi$ and $\sin 3\phi$) and feeds eight residual blocks with dilations from 1 to 128, in the manner of temporal convolutional networks [36]. Each block is modulated by feature-wise linear modulation (FiLM) [37] from a style vector that a small perceptron computes from c and eight noise values. A head pools the trunk to 1 s knots and predicts, at each knot, six non-negative harmonic weights

normalised to sum to one, their phases, a baseline and a bounded gain; the knots are interpolated back to 40 Hz. The output is

$$y(t) = G(t) \sum_{h=1}^{6} w_h(t)\, \sin\big(h\,\phi(t) + \theta_h(t)\big) + b(t) + 0.15 \tanh \rho(t) + m(t), \quad (6)$$

where $\rho(t)$ is a free residual from the trunk; its bound keeps it from overriding the rhythm. Three further details make the beats more physiological. Systole keeps a fixed duration (0.35 of the mean interval) and diastole absorbs the interval changes, so the peak-to-peak intervals keep the commanded dispersion. Each pulse is scaled by $(\tau_{k-1} f)^{\beta}$, with $\beta \in [0, 1]$ learned per condition, which mimics the dependence of pulse amplitude on the preceding filling time. A periodic motion term $m(t)$ with a learned cadence of 0.5–3.5 Hz and three harmonics represents wrist motion as a waveform; it is not driven by an accelerometer. The signal is generated with 240 extra samples on each side, band-limited by the data-view filter, cropped to the central 30 s and min–max normalised.

The critic examines the signal x together with its scaled differences $10\Delta x$ and $20\Delta^2 x$ at three time scales with spectral normalisation [38] and a projection term on c [39]. We trained with the hinge loss [40], adding a term that treats real windows paired with a mismatched condition (another dataset, or a known rate more than 10 bpm away) as negatives, and with lazy R1 regularisation [41, 42] every 16 steps with weight 0.1. The generator also minimises a distribution loss that matches the sorted, normalised log power spectra (blocks of 4, 10 and 20 s) and autocorrelation functions (lags up to 5 s) of generated and real batches, both pooled and within each dataset. Optimisation used Adam [43] with $\beta = (0, 0.99)$, a learning rate of 10^{-4} decayed by a cosine schedule to 10^{-5}, batches of 32, 20,000 updates, gradient clipping and an exponential moving average of the generator weights with decay 0.995 [44].

3.4. PhysioFlow: conditional rectified flow

PhysioFlow (Fig. 3b) replaces the adversarial game by rectified flow [45], a form of flow matching [46]. Let x_0 be a real window standardised to zero mean and unit variance, $\varepsilon \sim \mathcal{N}(0, I)$ and $t \sim \mathcal{U}(0, 1)$. A network v_θ is trained on the straight path $x_t = (1 - t)\, x_0 + t\, \varepsilon$ to predict the velocity $\varepsilon - x_0$:

$$\begin{aligned} \mathcal{L} &= \mathbb{E}\, \big\| v_\theta(x_t, t, s, c) - (\varepsilon - x_0) \big\|^2 + \lambda\, \mathbb{E}\, \Big[(1 - t)^2\, \big\| 10\, (\Delta \hat{x}_0 - \Delta x_0) \big\|^2 \Big], \\ \hat{x}_0 &= x_t - t\, v_\theta(x_t, t, s, c), \end{aligned} \quad (7)$$

where s is the scaffold and the second term ($\lambda = 1$) penalises errors in the first difference of the implied clean signal, weighted towards low noise levels. The network is a one-dimensional U-Net [47] with five resolutions (1200 down to 75 samples), 32 to 160 channels, two residual blocks per level, self-attention at 150 and 75 samples and FiLM modulation from time and condition embeddings; it has 6,923,137 parameters. Its input concatenates x_t with three scaffold channels, $[\sin\phi, \cos\phi, \text{flag}]$. The scaffold is supplied only when the condition marks the rate as known, as in training, where windows with masked labels were learned without it; a further 15% of training windows drop the scaffold and 10% drop the condition, which would allow classifier-free guidance [48]. Sampling integrates $\mathrm{d}x/\mathrm{d}t = v_\theta$ from $t = 1$ to 0 with 16 steps of Heun's method [49] (31 network evaluations) without guidance and min–max normalises the result. PhysioFlow has no critic and no output filter. Training used AdamW with a learning rate of 3×10^{-4} and a cosine schedule, batches of 128, 20,000 updates, gradient clipping at norm 1 and an exponential moving average of the weights.

3.5. Baselines and variants

Three references accompany the two generators. The *Gaussian* baseline is the parametric pulse of Eq. (3). *LSM-GAN* is the generator previously deployed in the simulator, a log-spectral-matching GAN [22] with 117,940 parameters that maps noise to a 30 s sequence; it was trained on BIDMC in a 0.9–5 Hz band and has no condition input. *Real (other subjects)* scores real windows of training subjects in place of a generator, which shows the level that a different real person reaches under each metric. Two variants appear in the development history: *PhysioFlow-SB* adds a loss on the log power spectrum in the stop bands of the data view (weight 0.1), and the retrained models PhysioMix-G, PhysioFlow-G, -D and -E are described in Section 4.6. Fig. 4 shows strips of the three generators next to real test windows, and Table 1 gives their size and speed.

4. Evaluation protocol

4.1. Datasets and splits

Table 2 lists the training data. The six datasets come from four public sources [51, 52, 8, 53, 54, 55, 56]; all recordings are de-identified, and no new data were collected from human participants for this study. They cover healthy adults at rest and after exercise (FINGER), critically ill adults

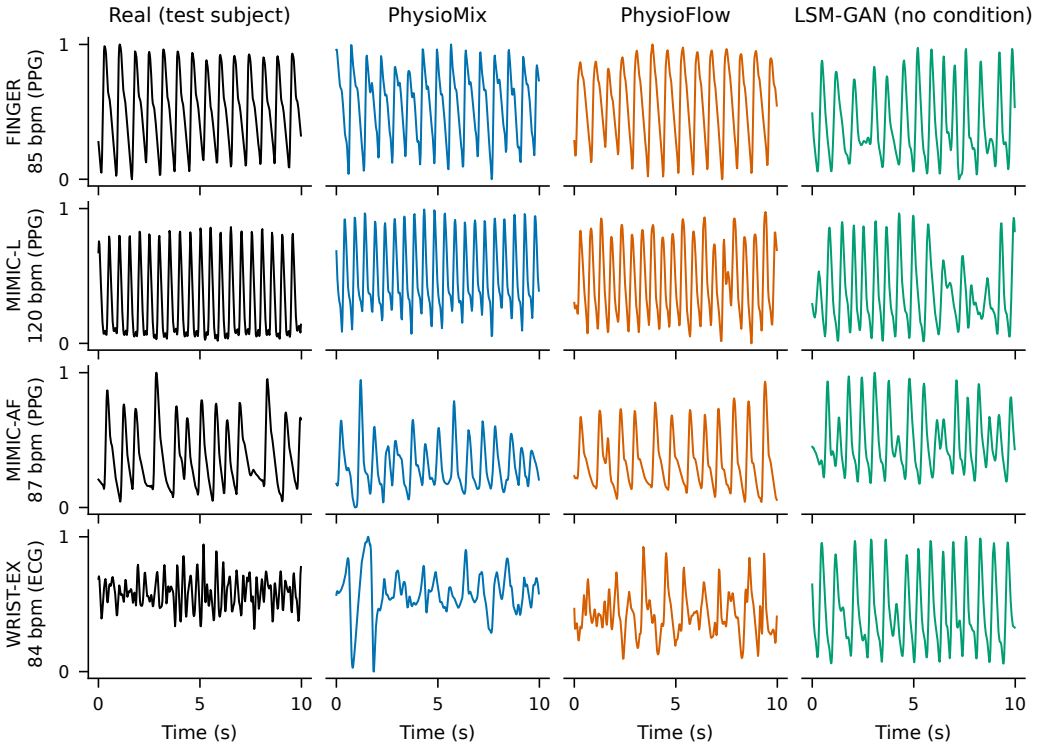


Figure 4: Ten-second excerpts of 30 s strips. The left column shows a randomly drawn test window of each dataset; PhysioMix and PhysioFlow were prompted with that window's dataset, its rate (PPG-derived, or ECG-derived for WRIST-EX) and the dataset's default irregularity, with the same noise seed for every row. LSM-GAN takes no condition and was given a different seed per row.

Table 1: Generator footprint. Latency is the median of seven generations of one 30 s strip after a warm-up, on one core of an AMD Ryzen 7 8840HS laptop with PyTorch 2.10.0 [50].

Generator	Parameters	Type	Latency (ms)
PhysioMix	162,216	GAN, constructed rhythm	5.2
PhysioFlow	6,923,137	Rectified flow, 16 Heun steps	716
LSM-GAN	117,940	GAN, unconditional	2.9

(BIDMC and MIMIC-L, the latter derived from MIMIC-III [57]) including patients in atrial fibrillation (MIMIC-AF) and in sinus rhythm (MIMIC-nonAF), and healthy adults exercising with a wrist sensor (WRIST-EX). Each dataset was split by subject with a fixed seed, so no subject contributes to more than one split [30]. For MIMIC-L, subjects are waveform record groups, because unique patient identities could not be verified; known aliases between the AF records and the larger set were joined before splitting, and possible overlap between BIDMC and MIMIC-derived records was handled by excluding matching record identifiers. Windows whose first or last 2 s exceeded the middle 1–99% range by more than 50% were removed from every split. The FINGER infrared channel is inverted by the recording device and was negated; 433 of 444 records showed a consistent pulse delay after the ECG R peak, and the remaining 11 were excluded. The prepared data pack carries a SHA-256 digest that every training and evaluation run checked.

4.2. Locked validation criterion

Checkpoint and model selection used one criterion. It was frozen before the third development round, identified by its SHA-256 digest and not changed afterwards. For each validation pool d, generated windows are prompted with condition vectors drawn from the training windows of that dataset and compared with up to 256 subject-balanced validation windows:

$$\begin{aligned} Q_d = {} & \mathrm{MMD}^2_{\mathrm{ACF}} + \mathrm{MMD}^2_{\mathrm{PSD}} + \mathrm{MMD}^2_{\mathrm{morph}} \\ & + 0.5\,(1 - \mathrm{recall}_d) + 0.25\,(1 - \mathrm{precision}_d). \end{aligned} \tag{8}$$

The three squared maximum mean discrepancies [58] use Gaussian kernels with median-heuristic bandwidths on autocorrelation functions, log power spectra and nine morphology descriptors: rate, IBI CV, pulse width at half height, rise time, standard deviation, skewness, mean absolute slope, mean absolute curvature and the fraction of power above 5 Hz. Precision and recall are k-nearest-neighbour manifold estimates ($k = 5$) [59] in the same standardised descriptor space. The pools are weighted 0.35 (FINGER), 0.25 (MIMIC-L), 0.15 (BIDMC), 0.15 (MIMIC-AF and MIMIC-nonAF together) and 0.10 (WRIST-EX), and

$$S_{\mathrm{real}} = \sum_d w_d\, Q_d + 0.25\,|\ln \varrho|, \tag{9}$$

where ϱ is the ratio of the median pairwise distance among generated windows to that among domain-matched real training windows; the term penalises

Table 2: Training datasets and subject-disjoint splits. Cells give subjects / 30 s windows. MIMIC-L subjects are waveform record groups.

Dataset	Source	Recording	f_s (Hz)	Train	Validation	Test
FINGER	Li et al. [51]	Finger, handheld device; healthy adults at rest and seated after exercise	250	102 / 1,759	22 / 387	22 / 388
MIMIC-L	MIMIC PERform [8]	ICU pulse oximeter; critically ill adults (MIMIC-III)	125	358 / 2,706	76 / 585	240 / 1,836
BIDMC	BIDMC [54]	ICU pulse oximeter; critically ill adults	125	32 / 619	7 / 111	7 / 111
MIMIC-AF	MIMIC PERform [8]	ICU pulse oximeter; atrial fibrillation	125	12 / 469	2 / 78	2 / 80
MIMIC-nonAF	MIMIC PERform [8]	ICU pulse oximeter; sinus-rhythm controls	125	11 / 437	1 / 40	1 / 40
WRIST-EX	Jarchi and Casson [56]	Wrist, research device; healthy adults walking, running and cycling	256	4 / 83	2 / 89	2 / 30

collapsed or over-dispersed output. Controllability is scored on a grid of rates from 40 to 180 bpm in steps of 10 bpm, with eight samples per point and dataset and a peak-based rate estimator that is independent of the generator:

$$C = 0.5 \min\left(\frac{\mathrm{MAE}_{\mathrm{in}}}{10}, 2\right) + 0.25 \min\left(\frac{\mathrm{MAE}_{\mathrm{ext}}}{10}, 2\right) + 0.25\,(1 - \rho_S) + 0.25\,(1 - \mathrm{AUC}_{\mathrm{irr}}). \tag{10}$$

Here $\mathrm{MAE}_{\mathrm{in}}$ and $\mathrm{MAE}_{\mathrm{ext}}$ are the mean absolute rate errors in bpm inside and outside 60–140 bpm (an undetected rate counts as 20 bpm), ρ_S is the Spearman correlation between commanded and median measured rate, and $\mathrm{AUC}_{\mathrm{irr}}$ is the area under the curve for separating irregular ($\sigma = 0.20$) from regular ($\sigma = 0.02$) prompts by the root mean square of successive interval differences [60]. The score of a trained model adds a classifier two-sample test (C2ST) [61]:

$$S_{\mathrm{arm}} = S_{\mathrm{real}} + 0.5\,C + 0.5 \max\left(0, \mathrm{AUC}_{\mathrm{C2ST}} - 0.5\right), \tag{11}$$

where $\mathrm{AUC}_{\mathrm{C2ST}}$ is the validation AUC of a small convolutional probe (three strided convolutions with 16, 32 and 32 channels and about 11 k parameters) that sees the window and its first two differences. Lower values are better for Q_d, S_{real}, C and S_{arm}. A model also had to pass gates: finite output, no flat windows, no non-finite losses, critic gradient clipping in at most 1% of updates, $\varrho \in [0.75, 1.25]$, numerical agreement between the exported TorchScript and the checkpoint, and single-thread latency below 50 ms (GAN) or 5 s (flow) per 30 s window.

4.3. Pre-registered decision rules

Each development round followed the same procedure. A rule naming the candidate, the incumbent and the conditions for replacing the incumbent was written to a file, hashed and committed to version control before training. Candidates were trained with three seeds (42, 123 and 2026) on two NVIDIA T4 GPUs, and checkpoints were chosen every 500 (GAN) or 2000 (flow) updates on validation $S_{\mathrm{real}} + 0.5C$ computed with the averaged weights. The general rule replaced the incumbent only if every seed passed the gates, the mean S_{arm} was lower by more than the pooled seed standard deviation, and the mean S_{real} was not worse by more than 0.05. Round 9 targeted the wrist dataset instead: it required a wrist Q better than the incumbent's by more than the pooled standard deviation while S_{arm} stayed within one pooled

standard deviation of the incumbent. Because round 9 had been accepted for its wrist gain, later rules added a wrist guard: the mean wrist Q could not be worse than the incumbent's by more than the pooled standard deviation. The deployed seed was the candidate's lowest-S_{arm} seed. The FINGER test subjects were sealed until round 3, scored once, and afterwards reported without taking part in any decision.

4.4. Realism on unseen subjects

The development criterion pools datasets and was used for selection. To report realism per dataset, we scored the windows of subjects never used for training (validation and test splits) after all decisions had been made, prompting the generators with the conditions of these real windows. *Fidelity* is the share of generated windows that fall inside the k-nearest-neighbour manifold ($k = 5$) of the real windows in the standardised descriptor space, and *coverage* is the share of real windows that fall inside the generated manifold [59, 62]. For the C2ST, the derivative probe was trained and tested with subjects grouped into folds (four folds; two for MIMIC-nonAF), and the result is reported as

$$\text{indistinguishable} = 100\,\big(1 - |2\,\mathrm{AUC} - 1|\big)\ \%, \tag{12}$$

so that 100% means the probe performs at chance and 0% that it separates the two sets perfectly. Every metric was also computed for real windows of training subjects in place of a generator. Because the validation subjects had guided checkpoint selection, the per-dataset scores of the learned models are slightly optimistic.

4.5. Recording context and memorisation

A seven-class convolutional classifier trained only on real training windows recognises the recording context: BIDMC, MIMIC-AF, MIMIC-nonAF, MIMIC-L, FINGER at rest, FINGER after exercise and WRIST-EX (chance level 14%). We generated windows for each context and report the balanced accuracy with which the classifier recovers the prompted context, together with a swap test in which the same noise is prompted with all seven contexts. Memorisation was screened with the maximum normalised cross-correlation, over all shifts, between each generated window and the training windows of the same dataset.

4.6. Everyday wrist PPG

WRIST-EX contains four training subjects recorded with a research device during exercise. To test transfer to a consumer wearable, we used GalaxyPPG [63]: 24 participants who wore an Empatica E4 wrist sensor (blood-volume pulse at 64 Hz) during rest, cognitive tasks, walking and running, with a Polar H10 chest strap as rate reference. Windows were cut into the same data view; the median difference between the window PPG rate and the Polar rate was 1.2 bpm. In a fresh test (step A), with a rule fixed before scoring, the deployed models were prompted with the Polar rates of one half of the participants and compared with the other half, over three random halvings of the 24 participants. Three retraining steps followed, each with a rule fixed in code before training. Step B added 12 GalaxyPPG participants to the wrist dataset and judged a seed-42 pilot on the other 12, who were never trained on, and on the original validation guards. Step D gave everyday wrist PPG its own dataset code and moved the decision to four validation participants taken from the 12 training participants. Step E repeated step D with six validation participants, five halvings and three seeds. The 12 held-out participants were scored after each decision had been saved. They had entered the decisions of steps A and B, neither of which selected a model, so they are held out from training but not untouched.

4.7. Independent judge

We designed the derivative probe ourselves, so it might reward our own design choices. We therefore added a judge we did not train: PaPaGei-S [64], a frozen PPG foundation model with 5.8 M parameters pre-trained on more than 57,000 hours of public PPG. Windows were resampled from 40 to 125 Hz, split into three 10 s segments, standardised and embedded; a linear C2ST on the mean 512-dimensional embedding, grouped by subject, gave a second indistinguishability score. We also trained the derivative probe on band-limited copies of real and generated windows (0.5–1.5, 1.5–3, 3–5 and 5–8 Hz) to locate the cues that betray a generator. None of these results fed back into training or selection.

4.8. Measurements on the hardware

On the Raspberry Pi 4 (Cortex-A72, Python 3.12.3, PyTorch 2.5.1), the engine counted, during 30 s playbacks of an LSM-GAN strip, the samples generated, the deadlines missed by the writer (overruns), the samples dropped

Table 3: Converter output during 30 s playbacks on the Raspberry Pi 4. Counters come from the signal engine; the analogue output was not measured with an oscilloscope.

Configuration	Rate (Hz)	Generated	Overruns	Dropped	Underruns	Write errors
Headless, fixed sleep (initial)	1000	30,000	713	802	9	0
Headless, deadline sleep	1000	30,000	0	0	9	0
GUI + BLE + RX	1000	30,000	330	1169	9	0
GUI + BLE + RX, lighter redraw	1000	30,000	187	729	9	0
GUI + BLE + RX, lighter redraw	500	15,000	0	0	5	0

because the buffer was full, the reads that found the buffer empty (underruns) and the converter write errors. The receiver logged A0 before, during and after playback. Generator latency was measured on a laptop as described in Table 1.

5. Results

5.1. Real-time output on the Raspberry Pi

Table 3 summarises the timing runs. In the first headless run at 1 kHz, a fixed sleep after the two I^2C writes made the writer fall behind: it missed 713 deadlines and dropped 802 samples in 30 s. Sleeping only until the next deadline removed both. With the interface, the Bluetooth server and the receiver running, 1 kHz still dropped 1,169 samples; skipping redraws of hidden pages and limiting the generator plot to 10 frames per second reduced the loss to 729. At 500 Hz the same configuration generated 15,000 samples with no overrun, no dropped sample and no write error. The five underruns were ticks at which the writer found the buffer momentarily empty; no generated sample was dropped. We therefore made 500 Hz the default on the Pi. The receiver sampled A0 at 99.995–100.002 Hz in all phases of the capture. These counters describe the software only; we did not measure the analogue output with an oscilloscope.

The PyTorch 2.10.0 wheel for aarch64 stopped with an illegal-instruction signal on the Cortex-A72, at an atomic instruction of the ARMv8.1 Large System Extensions that this ARMv8.0 core does not implement. PyTorch 2.5.1 ran correctly. On the Pi, the LSM-GAN TorchScript produced the same strip as on the laptop to within 2.4×10^{-8} V after the voltage mapping, and a cold start (import, load and generation) took 5.1 s. The deployed generators were not timed on the Pi.

The optical loopback did not reproduce the waveform. During a 30 s playback with only the infrared receiver connected, A0 stayed at the level

recorded with the LEDs off for the first 22.5 s and then jumped to a plateau near 3,300 counts. Over lags of ±250 ms, the largest absolute correlation between the detrended commanded and received signals was 0.045. We have not yet isolated the cause, which could lie in LED and receiver alignment, ambient light, grounding or the driver stage, so the optical path remains unvalidated.

5.2. Rate and rhythm control

Both generators followed the rate command (Table 4). Over three seeds, PhysioMix had a mean absolute rate error of 0.43 bpm between 60 and 140 bpm and 0.62 bpm at the extrapolated rates (40–50 and 150–180 bpm); PhysioFlow had 0.30 and 0.39 bpm. The Spearman correlation between commanded and measured rate was at least 0.999. Irregular prompts produced irregular rhythms: for a command of $\sigma = 0.20$, the measured IBI CV was 0.193 (PhysioMix) and 0.211 (PhysioFlow), and irregular and regular prompts were separated with an AUC of 0.967 and 0.998. A new measurement of the deployed models on our laptop, with near-regular prompts in all six contexts (Fig. 5), gave median rates within 0.17 bpm of the command for PhysioMix and within 2.2 bpm for PhysioFlow at every grid point; the largest PhysioFlow deviation occurred at 40 bpm in the wrist-exercise context. LSM-GAN has no rate input; on GalaxyPPG its median rate error was 14.4 bpm.

5.3. Development under pre-registered rules

Table 5 lists the decisions. Across rounds 3 to 7, PhysioMix improved from $S_{\text{arm}} = 0.596 \pm 0.026$ to 0.510 ± 0.053. Two kinds of change produced this progress. Generating a longer signal and cropping it (round 5) removed an artefact of circular filtering that the probe had learned to detect. The other changes concerned the critic: in round 5 every lazy R1 step hit the gradient-clipping limit, which weakened the penalty about tenfold; round 6 rescaled the limit, and round 7 reduced the R1 weight to 0.1. Round 9 added the motion term and the per-dataset distribution loss. It was accepted for its wrist gain (wrist Q 0.527 versus 0.653), while its S_{arm} of 0.530 stayed within the allowed band. PhysioFlow reached a lower S_{arm} (0.482 ± 0.015) and a lower probe AUC (0.703 versus 0.827), and its S_{real} was within the non-inferiority margin, but its wrist Q of 0.793 broke the wrist guard. It was therefore not deployed as the default and is offered in the application as a research mode. PhysioFlow-SB lowered S_{arm} further (0.445 ± 0.004) and

Table 4: Validation results of the deployed PhysioMix, the PhysioFlow research mode and the PhysioFlow-SB candidate (mean ± standard deviation over three seeds). The first five columns are selection quantities (lower is better); the remaining columns are components of C (Eq. 10): mean absolute rate error in bpm inside and outside 60–140 bpm, Spearman correlation, AUC for irregular versus regular prompts, and measured IBI CV for a command of 0.20.

Model	S_{arm}	S_{real}	C	AUC_{C2ST}	Wrist Q	MAE_{in}	MAE_{ext}	ρ_S	AUC_{irr}	CV at 0.20
PhysioMix	0.530 ± 0.029	0.344 ± 0.014	0.046 ± 0.014	0.827 ± 0.020	0.527 ± 0.060	0.43	0.62	0.999	0.967	0.193
PhysioFlow	0.482 ± 0.015	0.368 ± 0.011	0.025 ± 0.003	0.703 ± 0.022	0.793 ± 0.087	0.30	0.39	1.000	0.998	0.211
PhysioFlow-SB	0.445 ± 0.004	0.321 ± 0.010	0.039 ± 0.024	0.709 ± 0.015	0.597 ± 0.027	0.28	0.95	0.997	0.999	0.210

Table 5: Decisions taken under rules fixed before training (or, for step A, before scoring). Gaps and margins in steps A–E are in percentage points of indistinguishability.

Step	Candidate	Data used	Key comparison	Outcome
R9	PhysioMix R9 vs. R7	Validation	wrist Q 0.527 vs. 0.653 (gap $0.126 > 0.059$); S_{arm} $0.530 \leq 0.510 + 0.043$	Deployed (seed 123)
R10	PhysioFlow vs. PhysioMix R9	Validation	S_{arm} gap $0.048 > 0.023$; wrist Q $0.793 > 0.527 + 0.074$	Not deployed; research mode
L6	PhysioFlow-SB vs. PhysioMix R9	Validation	S_{arm} gap $0.085 > 0.021$; wrist Q $0.597 > 0.573$	Not deployed
A	PhysioFlow vs. PhysioMix	GalaxyPPG, 24 subjects	indistinguishable gap $-0.04 < 2.01$; Q worse by more than 0.05	PhysioMix kept
B	PhysioFlow-G, PhysioMix-G	GalaxyPPG, 12 held-out subjects; validation	PhysioFlow-G gap $31.1 > 7.1$ but wrist Q $0.950 > 0.587$; PhysioMix-G gap $2.6 < 4.0$	No final run
D	PhysioFlow-D, PhysioMix-D	4 everyday-wrist validation subjects; validation	PhysioFlow-D gap $28.8 < 32.8$; PhysioMix-D gap $6.7 < 11.4$	No final run
E	PhysioFlow-E, 3 seeds	6 everyday-wrist validation subjects; validation	gap $25.6 < 32.5$; Q, rate and validation guards met	PhysioMix kept

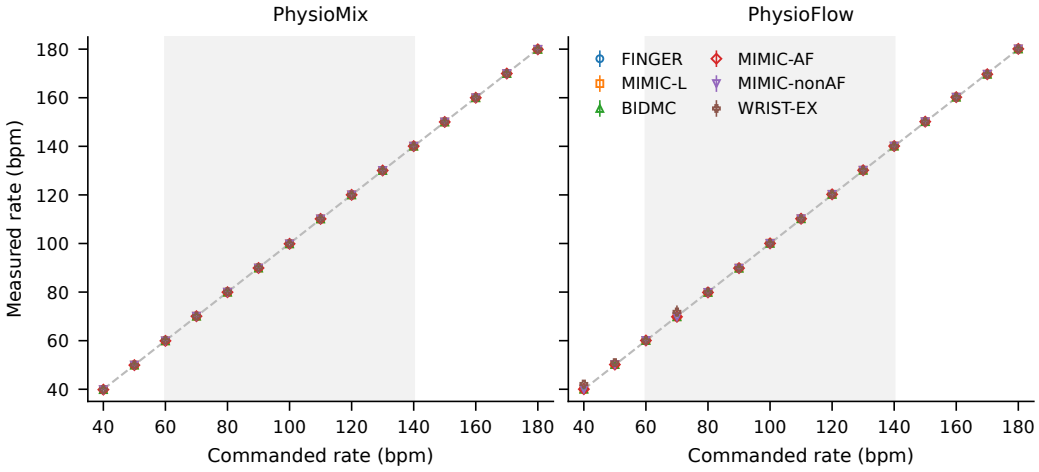


Figure 5: Measured versus commanded rate for the deployed generators in the six recording contexts (median and interquartile range of eight strips per point, IBI CV command 0.02; laptop measurement). The shaded band marks 60–140 bpm; the training labels cover 40–135 bpm.

narrowed the wrist gap (0.597), but it again missed the guard, whose limit was 0.573.

5.4. Realism on unseen subjects

Table 6 and Fig. 6 report realism per dataset. Both learned models placed most windows inside the real manifold on FINGER and MIMIC-L (fidelity 95–98%), but they differed in coverage and in how easily the probe found them. On FINGER, MIMIC-L, BIDMC and MIMIC-AF, PhysioFlow covered 84–95% of the real windows and was 50–91% indistinguishable; PhysioMix covered 46–76% and was 7–45% indistinguishable. Real windows of

Table 6: Realism on subjects never used for training. Each cell gives fidelity / coverage / indistinguishable, all in %. n is the number of unseen subjects. Real (other subjects) replaces the generator with real windows of training subjects.

Dataset	n	PhysioMix	PhysioFlow	LSM-GAN	Gaussian	Real (other subjects)
FINGER	44	95 / 76 / 9	96 / 92 / 50	12 / 0 / 0	1 / 1 / 0	98 / 99 / 94
MIMIC-L	191	98 / 63 / 36	97 / 88 / 91	97 / 1 / 0	70 / 0 / 0	98 / 95 / 97
BIDMC	14	49 / 67 / 45	57 / 84 / 67	79 / 2 / 0	0 / 60 / 0	51 / 88 / 96
MIMIC-AF	4	51 / 46 / 7	57 / 95 / 82	27 / 1 / 0	1 / 0 / 0	55 / 83 / 56
MIMIC-nonAF	2	10 / 94 / 80	11 / 41 / 39	5 / 0 / 0	0 / 0 / 0	11 / 18 / 59
WRIST-EX	4	92 / 56 / 52	65 / 61 / 52	0 / 0 / 0	0 / 0 / 0	77 / 54 / 92

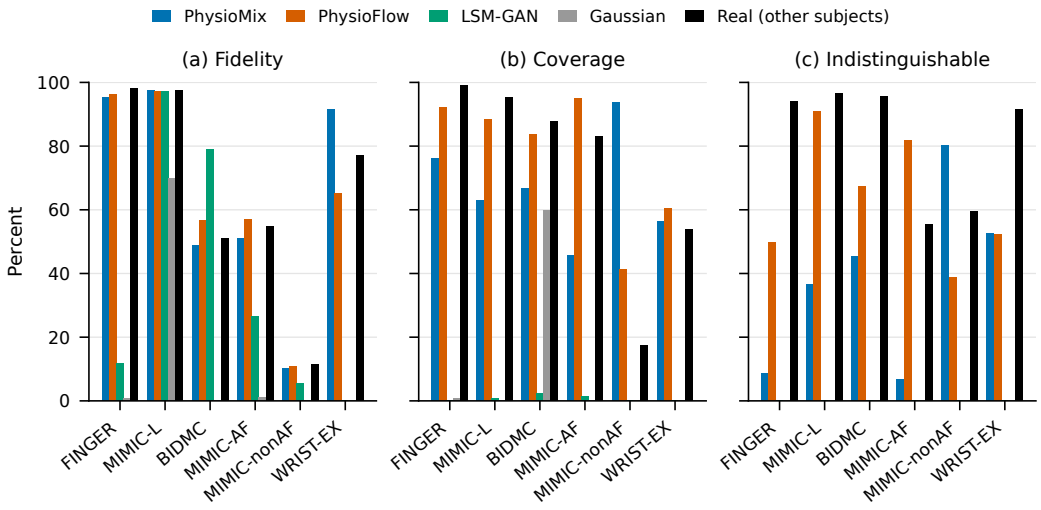


Figure 6: Realism on subjects never used for training, by dataset and source (values of Table 6).

other subjects reached 94–97% on FINGER, MIMIC-L and BIDMC, and PhysioFlow came closest to that level on MIMIC-L (91% against 97%). The order reversed on MIMIC-nonAF, where PhysioMix covered 94% and was 80% indistinguishable, but this dataset has two unseen subjects and real windows of other subjects reached only 59%. On WRIST-EX the two models were equally detectable (52%), and PhysioMix had the higher fidelity (92% against 65%). LSM-GAN produced plausible pulses on MIMIC-L and BIDMC (fidelity 97% and 79%) but covered at most 2% of any dataset and was never indistinguishable, which is consistent with a collapse onto one pulse type; scored in its own 0.9–5 Hz band on BIDMC, it reached 37%. The Gaussian pulse was 0% indistinguishable everywhere.

5.5. Recording context and memorisation

The context classifier recovered the prompted context for 54% of PhysioMix windows and 68% of PhysioFlow windows (balanced accuracy), against 51% for real windows of unseen subjects (Table 7). When only the condition changed, 49% (PhysioMix) and 71% (PhysioFlow) of the answers followed it; LSM-GAN, which has no condition, stayed at chance. A score above the real level is not better. It shows that PhysioFlow's windows are more typical of their context than real windows are, that is, less diverse. The wrist-exercise prompt was recognised in 77% (PhysioMix) and 94% (PhysioFlow)

Table 7: Recording-context following and memorisation screening (Real, unseen: real windows of subjects never used for training). Balanced accuracy of a seven-class context classifier trained on real windows (chance 14%), recognition of the wrist-exercise prompt, and the share of answers that follow the prompt when only the condition changes (swap). The last six columns give the median, over generated windows, of the maximum correlation with any training window of the same dataset.

	Context following (%)			Median maximum correlation with training windows					
Source	Balanced	Wrist-ex	Swap	FINGER	MIMIC-L	BIDMC	MIMIC-AF	MIMIC-nonAF	WRIST-EX
Real, unseen	51	86	–	0.75	0.87	0.85	0.41	0.82	0.28
PhysioMix	54	77	49	0.81	0.84	0.77	0.42	0.90	0.28
PhysioFlow	68	94	71	0.80	0.83	0.78	0.41	0.90	0.46
LSM-GAN	14	0	14	0.48	0.47	0.46	0.43	0.41	0.30
Gaussian	20	8	17	0.75	0.86	0.83	0.41	0.84	0.27

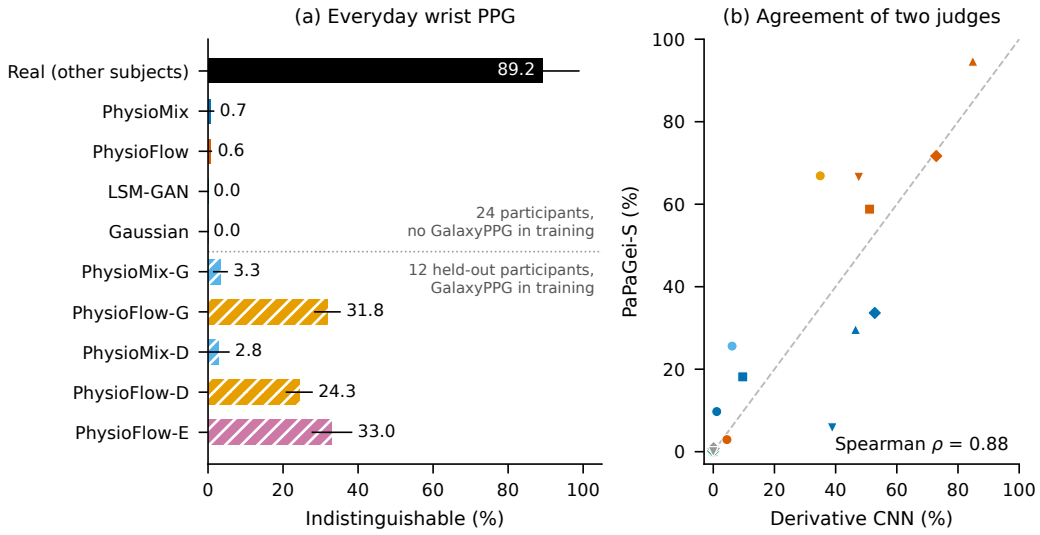


Figure 7: (a) Indistinguishability on everyday wrist PPG. Upper block: fresh test of models trained without GalaxyPPG (24 participants). Lower block: models retrained with GalaxyPPG, scored on 12 held-out participants (hatched). Error bars: standard deviation across halvings. (b) Indistinguishability according to the derivative probe and to the frozen PaPaGei-S foundation model for 22 generator–dataset pairs (real data excluded). Colours identify the generator as in (a); markers identify the dataset: circle, GalaxyPPG; square, FINGER; upward triangle, MIMIC-L; diamond, BIDMC; downward triangle, WRIST-EX.

of cases, against 86% for real wrist windows. The memorisation screen found no PhysioMix window with a correlation above 0.99 to a training window. For PhysioFlow, 0.8% of BIDMC windows and 1.6% of MIMIC-nonAF windows exceeded 0.99, which points to partial memorisation in the two smallest datasets. The median maximum correlations were otherwise close to those of real unseen windows, except for PhysioFlow on WRIST-EX (0.46 against 0.28).

5.6. Everyday wrist PPG and the independent judge

Neither deployed model transferred to everyday wrist PPG (Table 8a). Over three halvings of the 24 GalaxyPPG participants, real windows of other participants were 89% indistinguishable, PhysioMix 0.7% and PhysioFlow 0.6%; LSM-GAN and the Gaussian pulse were never indistinguishable. PhysioMix kept its rate accuracy (median error 1.5 bpm) and had the best distribution score Q among the generators (0.84). PhysioFlow did not meet the rule of step A, and PhysioMix remained the default.

Table 8: Everyday wrist PPG (GalaxyPPG, Empatica E4). (a) Fresh test of models trained without these data, over three halvings of all 24 participants. (b) Models retrained with GalaxyPPG, scored on the 12 participants held out from training; each row also gives the deployed PhysioMix and real windows of other participants from the same run. Indistinguishable is mean ± standard deviation across halvings (step E: mean over three seeds of the per-seed values). Rate error is the median absolute error against the prompted Polar rate.

(a)

Source	Indist. (%)	Fidelity (%)	Coverage (%)	Q	Rate error (bpm)	Within 5 bpm (%)
Real (other subjects)	89.2 ± 9.9	92	94	0.08	1.2	72
PhysioMix	0.7 ± 1.0	71	34	0.84	1.5	61
PhysioFlow	0.6 ± 0.4	41	7	1.33	1.7	70
LSM-GAN	0.0 ± 0.0	10	0	2.13	14.4	15
Gaussian	0.0 ± 0.0	19	0	2.53	0.0	100

(b)

		Indistinguishable (%)			Q			
Step	Model	Model	PhysioMix	Real	Model	PhysioMix	Rate error (bpm)	Within 5 bpm (%)
B	PhysioMix-G	3.3 ± 2.0	0.7	71.4	0.28	0.85	1.3	89
B	PhysioFlow-G	31.8 ± 3.6	0.7	71.4	0.19	0.85	1.2	89
D	PhysioMix-D	2.8 ± 3.0	0.7	71.4	0.49	0.85	1.4	84
D	PhysioFlow-D	24.3 ± 3.6	0.7	71.4	0.23	0.85	1.2	89
E	PhysioFlow-E (3 seeds)	33.0 ± 5.4	0.4	69.9	0.21	0.88	1.2	88

Training on everyday wrist data changed the picture for the flow model only (Table 8b and Fig. 7a). With 12 GalaxyPPG participants in the wrist dataset (step B), PhysioFlow-G reached 31.8% on the 12 held-out participants, against 0.7% for the deployed PhysioMix in the same run, but its wrist-exercise Q on the original validation rose to 0.950 and breached the regression guard. Giving everyday wrist PPG its own dataset code (step D) removed this regression (0.574) and gave 24.3% on the held-out participants. On the four validation participants, however, the gain over PhysioMix (28.8 points) fell short of the rule's margin of 32.8 points, twice the largest standard deviation across halvings. Step E used six validation participants and three seeds. The seeds agreed closely ($25.6 \pm 1.2\%$), but the standard deviation across halvings was 16.3 points, so the margin rose to 32.5 points and the gain of 25.6 points again fell short. On the held-out participants PhysioFlow-E reached 33.0% (seed range 31.9–34.4%) with a median rate error of 1.2 bpm. PhysioMix retrained on the same data improved little (2.8–3.3%).

PaPaGei-S ordered the generators much like our probe (Fig. 7b): over 22 generator–dataset pairs the Spearman correlation between the two indistinguishability scores was 0.88, and within datasets it ranged from 0.80 to 1.00. The foundation model was more lenient overall. On GalaxyPPG it rated PhysioMix 9.7% and PhysioFlow-G 66.9% indistinguishable, against 1.1% and 35.0% for the probe in the same run. The band-limited probes located what gives the deployed PhysioMix away on GalaxyPPG: it was 56%, 80% and 52% indistinguishable in the 0.5–1.5, 1.5–3 and 3–5 Hz bands, but 1% in the 5–8 Hz band, where real windows of other participants reached 89%.

5.7. Computational footprint

On one laptop core, PhysioMix generated a 30 s strip in 5.2 ms, PhysioFlow in 716 ms and LSM-GAN in 2.9 ms (Table 1). PhysioFlow is therefore about 140 times slower than PhysioMix and has 43 times more parameters. In the simulator it runs in the worker process and does not block the converter thread; its latency on the Pi has not been measured.

6. Discussion

6.1. Separating rhythm from waveform

Constructing the beat sequence and learning only the waveform around it gave a degree of rate control that the training data alone could not provide. The training labels cover 40–135 bpm, yet both models met the command

with mean errors below 1 bpm up to 180 bpm, because the scaffold sets the beat times and the learned part cannot move them. In PhysioMix the residual is bounded for this reason; in PhysioFlow the network is conditioned on the scaffold and was never asked to infer the rhythm when the scaffold was present. The same design limits what the models can learn about rhythm. The inter-beat intervals are log-normal and independent, so sequential patterns such as those of atrial fibrillation or ectopic beats are not reproduced and would need a rhythm model of their own [16]. PhysioFlow nevertheless covered 95% of the MIMIC-AF windows, which suggests that it learned the morphological changes that accompany an irregular rhythm even though the interval process was imposed.

6.2. GAN or rectified flow

On the finger and intensive-care datasets, rectified flow produced waveforms that the probe and the foundation model both found harder to separate from real ones, with higher coverage and closer context following. The development history explains part of the difference. Two of the GAN rounds were spent on the stability of adversarial training (Section 5.3), whereas the flow model has a regression loss and no adversary, and its improvements came from the data path (when to supply the scaffold) and from the distribution of training times. The probe AUC tells the same story: 0.70 for PhysioFlow against 0.83 for PhysioMix on validation, although the GAN's critic was itself shown first and second differences. The price is compute. PhysioFlow needs 31 network evaluations per strip and about 140 times the CPU time of PhysioMix. For a simulator that generates 30 s strips ahead of playback, this is acceptable on a laptop; on the Pi it remains to be measured. The flow model is also less diverse: it followed contexts more consistently than real data do, and it showed signs of memorisation in the two smallest datasets. Both observations argue for more training subjects rather than for a larger network.

6.3. Everyday wrist PPG and small validation sets

Both deployed models failed on GalaxyPPG, and the band-limited probes show why: PhysioMix reproduced the pulse bands but not the 5–8 Hz texture of the E4 signal. Realism on one wrist dataset recorded with a research device during exercise did not carry over to a consumer device in daily life. Motion and sensor texture are device-specific, and a simulator that targets wearables needs data from the class of device being tested. Training on the

new device's data helped the flow model considerably, from about 3% for the deployed PhysioFlow to 24–33% on the same held-out participants, and once these data had their own dataset code, the original datasets did not suffer.

The pre-registered rules did not let this result change the deployed model, and the reason deserves attention. With four or six validation participants, the standard deviation across random halvings (16 points) was dominated by which participants served as the real reference: real windows of other participants themselves scored between 23% and 100% across the five halvings of step E. A rule that compares candidate and incumbent within each halving would have had more power, because the halving effect is shared by both models. We did not change the rule after seeing the scores, and a paired rule would have to be tested on participants not yet used. In this case the rules probably cost us a better model for everyday wrist data. In others, such as round 10, they prevented a switch that the pooled score favoured but that would have undone the wrist improvement of round 9. We see this as the intended behaviour of pre-registration [31]: with validation data reused over many rounds, a decision taken after seeing the scores would be hard to defend [29].

6.4. What the simulator can and cannot be used for

The learned modes suit tests of beat detection, pulse-rate estimation and rhythm classification on waveforms whose rate and irregularity are known exactly, in a chosen recording context, and the exercise of optical receivers with realistic morphology once the optical path is validated. Robustness tests of denoising algorithms [6, 7] or of sensors under vibration [65] would also benefit from strips whose ground-truth rhythm is known. The learned modes do not suit SpO_2 testing. Each strip is a single normalised shape mapped to both channels with a fixed nominal ratio (Eq. 2), and our LEDs (625 and 875 nm) differ from the 660 and 940 nm pair that is common in pulse oximetry. The data view also removes the respiratory baseline, which the parametric mode still provides. Like other functional testers, the simulator does not establish SpO_2 accuracy [10].

6.5. Limitations

The optical path has not been validated: the one loopback capture did not reproduce the commanded waveform, and the converter timing was verified with software counters rather than an oscilloscope. The deployed generators were not timed on the Pi. Several unseen cohorts are small (four subjects

in MIMIC-AF and WRIST-EX, two in MIMIC-nonAF), so per-dataset estimates there are uncertain, and MIMIC-L subjects are record groups rather than verified patients. Validation data were reused over many development rounds, and the GalaxyPPG held-out participants had entered two earlier decisions. All metrics are defined on a normalised 0.5–8 Hz view without absolute amplitude or paired infrared and red channels; the FINGER dataset contains a red channel, which we stored but did not use. The labels are engineering labels, and nothing in this work is a clinical validation. The study comes from one laboratory and one prototype.

7. Conclusions

We built a low-cost PPG simulator on a Raspberry Pi 4 and two generators for it that construct the beat sequence and learn the waveform around it. This made rate control exact for both a GAN and a rectified-flow model, and the flow model was markedly harder to tell apart from real PPG on finger and intensive-care data. Neither model transferred to everyday wrist PPG from a new device without that device's data. With those data the flow model improved substantially, but not by the margin we had fixed in advance, and the pre-registered rules kept the deployed model unchanged. The next steps are to validate the optical path, to time the generators on the Pi, to learn paired infrared and red channels and to test a paired decision rule on participants not used so far.

CRediT authorship contribution statement

[First author]: Conceptualization, Methodology, Investigation, Software, Formal analysis, Writing – original draft. **[Second author]:** Software, Visualization, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All training and test datasets are public: FINGER [52], MIMIC PERform [53], BIDMC [54], wrist PPG during exercise [56] and GalaxyPPG (doi:10.5281/zenodo.14635823) [63]. Source code, board designs, trained generators with their SHA-256 digests, the frozen decision rules, the JSON records of the evaluation runs and the script that produces every figure and table of this paper from those records are available at `https://github.com/nguyenhuy0426/PPG_simulator`. The prepared data pack (49 MB) is not redistributed; the preparation code in the repository builds it from the public sources, and its SHA-256 digest is recorded so that a rebuilt pack can be checked.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to draft and edit the manuscript text and to write the script that draws every figure with Matplotlib (Fig. 2 arranges KiCad renders and an application screenshot) and generates the tables from the evaluation records. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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